that fluorescence in the crystals led to the emission of X-rays, which penetrated the wrapping paper. However, Becquerel soon found that the radiation causing the blackening was not "a transformation of solar energy" because it was found to occur even with assemblies that had not been exposed to light; the uranyl salt obviously produced radiation spontaneously. This radiation, which was first called uranium rays (or Becquerel rays) but later termed radioactive radiation (or simply *radioactivity*)[1], was similar to X-rays in that it ionized air, as observed through the discharge of electroscopes.

Marie Curie subsequently showed that all uranium and thorium compounds produced *ionizing radiation* independent of the chemical composition of the salts. This was convincing evidence that the radiation was a property of the element uranium or thorium. Moreover, she observed that some uranium minerals such as pitchblende produced more ionizing radiation than pure uranium compounds. She wrote: "this phenomenon leads to the assumption that these minerals contain elements which are more active than uranium". She and her husband, Pierre Curie, began a careful purification of pitchblende, measuring the amount of radiation in the solution and in the precipitate after each precipitation separation step. These first *radiochemical* investigations were highly successful: "while carrying out these operations, more active products are obtained. Finally, we obtained a substance whose activity was 400 times larger than that of uranium. We therefore believe that the substance that we have isolated from pitchblende is a hitherto unknown metal. If the existence of this metal can be affirmed, we suggest the name *polonium*." It was in the publication reporting the discovery of polonium in 1898 that the word radioactive was used for the first time. It may be noted that the same element was simultaneously and independently discovered by W. Marckwald who called it "radiotellurium".

In the same year the Curies, together with G. Bemont, isolated another radioactive substance for which they suggested the name *radium*. In order to prove that polonium and radium were in fact two new elements, large amounts of pitchblende were processed, and in 1902 M. Curie announced that she had been able to isolate about 0.1 g of pure radium chloride from more than one ton of pitchblende waste. The determination of the atomic weight of radium and the measurement of its emission spectrum provided the final proof that a new element had been isolated.

1.2. RADIOACTIVE DECAY

While investigating the radiochemical properties of uranium, W. Crookes and Becquerel made an important discovery. Precipitating a carbonate salt from a solution containing uranyl ions, they discovered that while the uranium remained in the supernatant liquid in the form of the soluble uranyl carbonate complex, the radioactivity originally associated

[1] The word radioactivity refers to the phenomenon *per se* as well as the intensity of the radiation observed.

with the uranium was now present in the precipitate, which contained no uranium. Moreover, the radioactivity of the precipitate slowly decreased with time, whereas the supernatant liquid showed a growth of radioactivity during the same period (Fig. 1.1). We know now that this measurement of radioactivity was concerned with only beta- and gamma-radiations, and not with the alpha-radiation which is emitted directly by uranium.

Similar results were obtained by E. Rutherford and F. Soddy when investigating the radioactivity of thorium. Later Rutherford and F. E. Dorn found that radioactive gases (*emanation*) could be separated from salts of uranium and thorium. After separation of the gas from the salt, the radioactivity of the gas decreased with time, while new radioactivity grew in the salt in a manner similar to that shown in Fig. 1.1. The rate of increase with time of the radioactivity in the salt was found to be completely independent of chemical processes, temperature, etc. Rutherford and Soddy concluded from these observations that radioactivity was due to changes within the atoms themselves. They proposed that, when radioactive decay occurred, the atoms of the original elements (e.g. of U or of Th) were transformed into atoms of new elements.

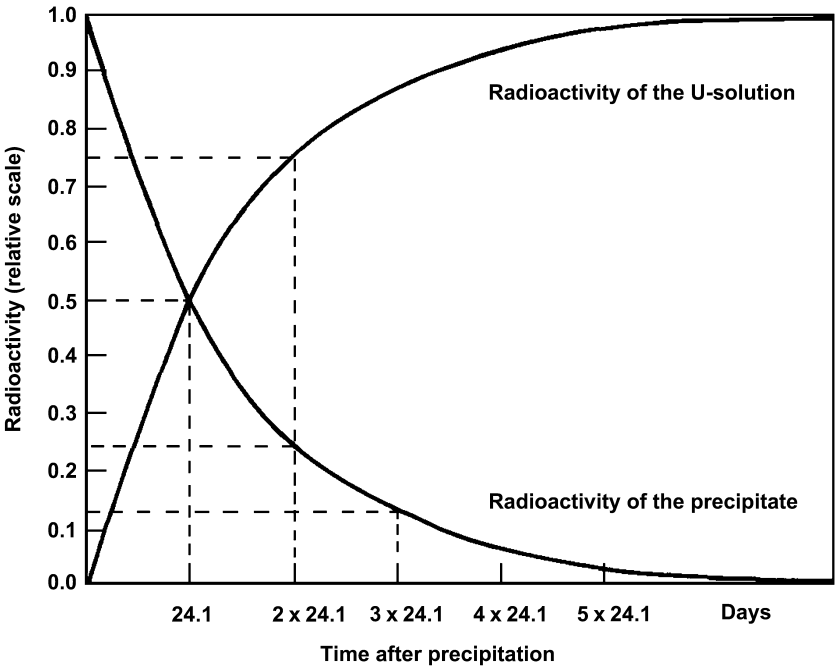

Figure 1.1 Measured change in radioactivity from carbonate precipitate and supernatant uranium solution, i.e. the separation of daughter element UX (Th) from parent radioelement uranium.

The radioactive elements were called *radioelements*. Lacking names for these radioelements, letters such as X, Y, Z, A, B, etc., were added to the symbol for the primary (i.e. parent) element. Thus, UX was produced from the radioactive decay of uranium, ThX from that of thorium, etc. These new radioelements (UX, ThX, etc.) had chemical properties that were different from the original elements, and could be separated from them through chemical processes such as precipitation, volatilization, electrolytic deposition, etc. The radioactive daughter elements decayed further to form still other elements, symbolized as UY, ThA, etc. A typical decay chain could be written: Ra → Rn → RaA → RaB → , etc.; see Fig. 1.2.

A careful study of the radiation emitted from these radioactive elements demonstrated that it consisted of three components which were given the designation alpha (α), beta (β), and gamma (γ). *Alpha-radiation* was shown to be identical to helium ions, whereas *beta-radiation* was identical to electrons. *Gamma-radiation* had the same electromagnetic nature as X-rays but was of higher energy. The rate of radioactive decay per unit weight was found to be fixed for any specific radioelement, no matter what its chemical or physical state was, though this rate differed greatly for different radioelements. The decay rate could be expressed in terms of a *half-life*, which is the time it takes for the radioactivity of a radioelement to decay to one-half of its original value. Half-lives for the different radioelements were found to vary from fractions of a second to millions of years; e.g. that of ThA is 0.1 of a second, of UX it is 24.1 days (Fig. 1.1), and of uranium, millions of years.

1.3. DISCOVERY OF ISOTOPES

By 1910 approximately 40 different chemical species had been identified through their chemical nature, the properties of their radiation, and their characteristic half-lives. The study of the generic relationships in the decay of the radioactive species showed that the radioelements could be divided into three distinct series. Two of these originated in uranium and the third in thorium. B. Boltwood found that all three of the series ended in the same element — lead.

A major difficulty obvious to scientists at that time involved the fact that while it was known from the Periodic Table (Appendix I) that there was space for only 11 elements between lead and uranium, approximately 40 radioelements were known in the decay series from uranium to lead. To add to the confusion was the fact that it was found that in many cases it was not possible to separate some of the radioelements from each other by normal chemical means. For example, the radioelement RaD was found to be chemically identical to lead. In a similar manner, spectrographic investigations of the radioelement ionium showed exactly the same spectral lines that had been found previously to be due to the element thorium.

In 1913 K. Fajans and Soddy independently provided the explanation for these seemingly contradictory conditions. They stated that by the radioactive α-decay a new